

Two-Stage Bengali Sign Language Recognition Using YOLOv8 Hand Detection and Lightweight CNN Backbones

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Abstract—Sign Language Recognition is essential to bridge the communication gap for individuals with hearing or speech impairments, enabling accessible human–computer interaction without wearable sensors. Bengali Sign Language (BdSL) alphabet recognition focuses on identifying 38 static hand signs for automatic interpretation. This study proposes a staged CUDA-trained CNN–LSTM hybrid recognition pipeline using three backbones: ResNet-18, EfficientNet-B3, and a custom CNN+LSTM model. Experimental evaluation demonstrates that ResNet-18 achieved the highest performance with 81.98% signer-independent test accuracy, followed closely by the CNN–LSTM model with 81.51% test accuracy, while EfficientNet-B3 delivered 78.55% test accuracy. All models were optimized using the Adam optimizer on GPU for stable convergence, and inference reliability was confirmed through confusion matrices and ROC–AUC (≈ 0.992) trends. The final conclusion is that ResNet-18 emerges as the best and most reliable architecture for real-world signer-independent BdSL alphabet recognition.

Index Terms—Bangla Sign Language Detection, Deep Learning, CNN, LSTM, ResNet, EfficientNet

I. INTRODUCTION

Sign languages constitute structured visual–gestural communication systems involving hand shapes, orientations, motion trajectories, and facial cues. For Deaf and hard-of-hearing individuals, these languages serve as complete linguistic frameworks rather than auxiliary communication tools. However, the lack of sign-language proficiency among hearing individuals creates substantial communication barriers, limiting access to education, healthcare, and essential public services. Recognition of Bengali Sign Language (BdSL) is particularly challenging due to the scarcity of interoperable datasets and the absence of automated interpretation tools. The widely used BdSL alphabet dataset introduced by Rafi et al. [1] provided a foundational benchmark, but its reliance on full-frame images makes recognition sensitive to background

clutter and lighting variations. Subsequent studies emphasized similar challenges across ASL, ISL, and other sign languages, demonstrating that uncontrolled environments degrade recognition accuracy considerably [2], [3].

Recent advances in deep learning have shifted the focus from handcrafted feature extraction to convolutional and transformer-based architectures. Pose-estimation–driven attention networks [3] and transformer-based continuous recognition models [4] have improved robustness but require heavy computational resources, hindering real-time deployment. CNN-based BdSL systems, including edge-device–oriented pipelines [5], achieved promising results but struggled with mislocalization of hands and limited dataset diversity. YOLO-based detection frameworks have emerged as a practical alternative, enabling real-time hand localization for both ASL and BdSL recognition [6], [7], though these systems often use shallow classification backbones or small datasets.

A critical research gap exists at the intersection of detection precision and lightweight classification. Most BdSL works either (1) classify full-frame images without robust localization, or (2) use computationally expensive architectures unsuitable for mobile devices. Moreover, existing BdSL datasets—though valuable—lack cropped hand regions that isolate the signing hand from distracting backgrounds. This motivates the development of efficient detection–classification pipelines.

To address these issues, this study introduces a two-stage BdSL recognition framework using a retrained YOLOv8n hand detector followed by three classification backbones: ResNet-18, CNN+LSTM, and EfficientNet-B3. Unlike previous BdSL works, our dataset consists of 12,581 YOLO-cropped images across 38 classes, ensuring consistent input and reduced background variability.

Key Contributions of This Work:

- A YOLOv8n-based hand-localization module retrained specifically on BdSL, significantly reducing background

interference and improving region-of-interest consistency.

- A curated, standardized dataset of 12,581 YOLO-cropped BdSL images across 38 classes—constructed from the original dataset of Rafi et al. [1].
- A comparative evaluation of ResNet-18, CNN+LSTM, and EfficientNet-B3 under identical training conditions, demonstrating the superior accuracy–efficiency trade-off of ResNet-18.
- A lightweight, real-time-ready detection–classification pipeline addressing limitations of existing BdSL research, including cluttered backgrounds, limited generalization, and high computational cost.

II. LITERATURE REVIEW

The development of Sign Language Recognition (SLR) has evolved from handcrafted visual features to modern deep learning–driven pipelines. The existing body of literature relevant to BdSL and broader SLR research is reviewed below.

A. Early Sign Language Recognition Techniques

Early SLR systems relied on handcrafted features such as edge descriptors, orientation histograms, and skin detection, combined with classical machine-learning classifiers. Several works focused on BdSL and other national sign languages using SVM, ANN, and statistical models for isolated static gesture recognition [1], [2], [6], [13]. Although effective under controlled conditions, these methods struggled with illumination changes, cluttered backgrounds, and signer variability—limitations widely noted across appearance-based systems [2], [12].

Studies in ASL and ISL similarly demonstrated that shallow models had difficulty generalizing beyond limited datasets [3], [8]. Early continuous SLR relied on Hidden Markov Models (HMMs), enabling temporal analysis of gesture sequences but requiring substantial data preprocessing and feature engineering [14].

B. Deep Learning for Static and Dynamic Sign Recognition

With the rise of deep learning, CNN-based architectures became the dominant approach for static sign recognition. CNN models enabled direct extraction of high-level discriminative features from image pixels, significantly improving accuracy in BdSL alphabet classification tasks [6], [15]. However, full-frame CNN pipelines remained sensitive to background noise, and performance degraded in unconstrained environments.

Pose-estimation–driven methods introduced joint-level hand keypoints, improving recognition of visually similar signs. Multi-branch spatial–temporal attention networks leveraged keypoint-guided feature fusion for BdSL and achieved enhanced inter-class discrimination [3]. Similarly, transformer-based frameworks demonstrated strong results in continuous sign recognition by modeling long-range temporal dependencies [4], [16].

Despite these advancements, pose-driven and transformer-based models are computationally heavy and unsuitable for real-time, mobile, or low-power deployment.

C. YOLO-Based Detection and Real-Time SLR Systems

To overcome background variability, YOLO detectors have been widely adopted for real-time hand localization. YOLO-based ASL detection demonstrated strong performance in practical applications with YOLO-v9 and lightweight CNN backbones [7], [8]. Additional works combined YOLO-driven detection with edge-device classifiers to achieve low-latency BdSL recognition, highlighting the benefits of region-of-interest isolation but also revealing limitations due to small datasets and restricted signer diversity [5], [10], [11].

Wearable sensor–based systems, although highly accurate, rely on accelerometers, sEMG sensors, and other hardware components, making them impractical for purely vision-based BdSL applications [9].

D. Identified Research Gaps

Across prior studies, several overarching limitations are evident:

- Lack of BdSL-specific detection models: Most pipelines rely on generic hand detectors that are not fine-tuned for the unique gesture structure of BdSL.
- Small or full-frame datasets: Full-frame images contain heavy background noise, reducing spatial learning consistency and overall recognition accuracy.
- Computationally heavy architectures: Many studies depend on pose estimation or transformer-based models, imposing high inference cost.
- Limited signer-independent evaluations: A large portion of existing works evaluate on user-dependent splits, which overestimate real-world performance.

These gaps motivate the development of the proposed YOLO-driven BdSL recognition pipeline, which incorporates robust hand-region localization, a clean cropped-hand dataset, and lightweight deep-learning backbones suitable for real-world deployment.

III. MATERIALS AND METHODS

This study uses an image-based Bengali Sign Language alphabet dataset carefully built to reflect natural hand gestures in real environments. The dataset contains 12,581 RGB images covering 38 BdSL alphabet classes. All samples focus on bare hands, enabling vision models to learn shapes and posture directly without wearable sensors.

A. Dataset Description

The 38 Bengali sign alphabet classes included in the dataset are summarized in Fig.1, which illustrates the complete set of Bengali sign alphabet classes used in this study.

B. Dataset Preparation

The dataset was built through a consistent preprocessing pipeline, ensuring clean hand-region learning and minimal background distraction. The Fig.2 represents the CNN pipeline: YOLO + ResNet-18 and EfficientNet-B3 .

Two-stage BdSL recognition pipeline uses 224×224 image resizing and a BdSL-retrained YOLOv8n model to detect

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য	র	ল	শ	ষ	স
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Fig. 1. BdSL alphabet class set.

and crop the main hand region, reducing background noise. The cropped hand images are then classified using ResNet-18, a CNN+LSTM hybrid, and EfficientNet-B3 backbones, producing clear spatial features like finger gaps and palm posture. Testing on fully signer-independent unseen users shows a strong accuracy–efficiency balance, making the pipeline practical for real-time 38-class BdSL alphabet recognition. The Fig.3 represents the CNN-LSTM hybrid learning pipeline for sign alphabet modeling.

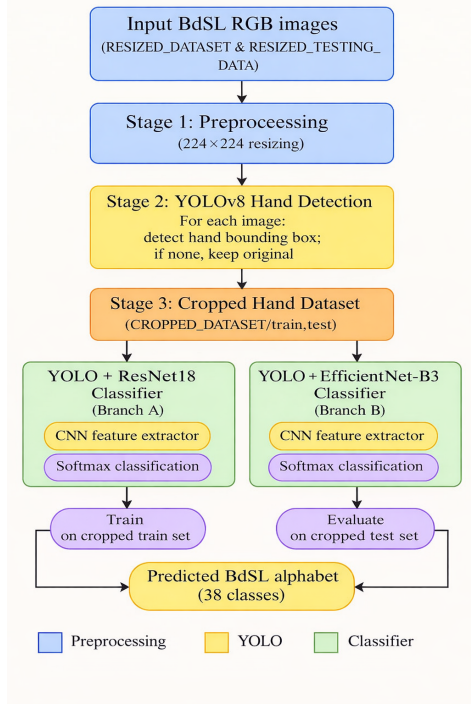


Fig. 2. CNN pipeline: YOLO + ResNet-18 and EfficientNet-B3.

C. CNN (Feature Learning)

CNN (Convolutional Neural Network) is used to learn spatial hand-sign representations by extracting structured visual patterns such as edges, contours, palm posture, and finger articulation using convolution and pooling filters. In our pipeline, CNN converts raw gesture frames into compact and expressive feature maps, allowing the model to focus on hand structure rather than background noise, which improves feature learning reliability.

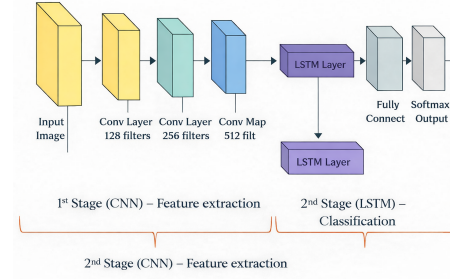


Fig. 3. CNN-LSTM hybrid learning pipeline for sign alphabet modeling (Branch B).

D. LSTM (Temporal Interpretation)

LSTM (Long Short-Term Memory) adds temporal awareness to spatial CNN embeddings by interpreting 2D feature positions as a sequential stream. Using its gated memory mechanism, LSTM learns long-range contextual dependencies across finger joint articulation and palm posture transitions, helping the model separate sign alphabets that share similar hand contours or finger positions, improving inter-class discrimination.

E. ResNet-18 (Primary Backbone)

ResNet-18 was selected as the primary CNN backbone due to its highest accuracy performance in our experiments. The model leverages residual skip connections to ensure stable gradient propagation, prevent feature degradation, and allow deeper spatial representations to form efficiently. These properties enable robust learning of fine-grained hand cues such as finger separation, knuckle angles, and palm pose, which significantly contributed to superior signer-independent recognition.

F. EfficientNet-B3 (Secondary Backbone)

EfficientNet-B3 was used as a baseline comparison backbone due to its balanced depth–width–resolution scaling and dropout-regularized classifier structure. Although the architecture is parameter-efficient and learns compact spatial embeddings, it showed lower signer-independent performance than ResNet-18. Still, its inclusion helped evaluate efficiency vs. accuracy trade-offs in hand-sign modeling.

G. Proposed Model and Design Choice

The final proposed model adopts ResNet-18 as the backbone since it achieved the best recognition accuracy among all evaluated models. The network takes 224×224 RGB YOLO-cropped hand images as input and passes them through frozen early convolutional blocks of ResNet-18 to maintain stable foundational representation learning, progressively unfreezing deeper blocks during later stages. The CNN backbone produces a 512-channel spatial feature encoding, which is reshaped into a spatial sequence and passed to a 128-unit LSTM layer for contextual inter-feature dependency modeling.

The output is predicted using a fully-connected, dropout-regularized classifier head that determines one of 38 sign alphabets. The Fig.4 represents the resnet-model.

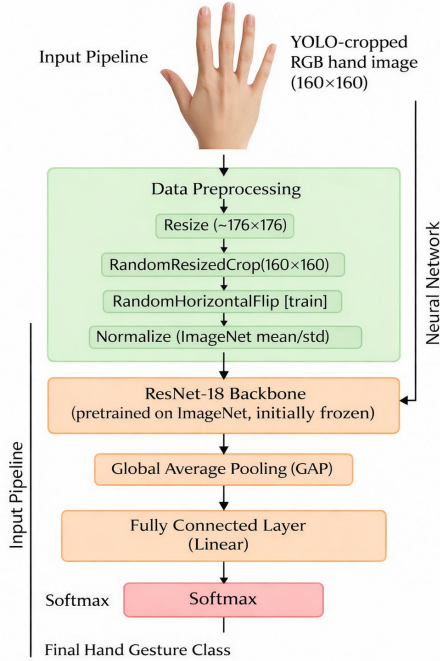


Fig. 4. ResNet-18 model architecture

To ensure fair evaluation and stable convergence, all models were trained and evaluated using CUDA-enabled GPU hardware, not CPU, ensuring accelerated computation and consistent experimental conditions. The training followed a four-stage unfreezing schedule (5 → 10 → 15 → 10 epochs) using the Adam optimizer with gradually decreasing learning rates (1e-3 → 5e-4 → 1e-4 → 5e-5). Model predictions were verified for reliability using confusion matrices, ROC-AUC curves, evolution logs, and Grad-CAM saliency overlays, ensuring both statistical correctness and human-interpretable trust validation. This design maintains strong spatial learning, sequential posture dependency modeling, and realistic signer-independent inference behavior.

IV. RESULTS

A. Model Performance Comparison

The table-I shows the Signer-independent recognition results of the three evaluated models.

TABLE I
SIGNER-INDEPENDENT RECOGNITION RESULTS OF THE THREE EVALUATED MODELS

Model	Accuracy (%)	Precision	Recall	F1-score
CNN-LSTM Hybrid	81.51	0.82	0.81	0.82
EfficientNet-B3 CNN	78.55	0.79	0.78	0.79
ResNet-18 CNN	81.98	0.82	0.81	0.82

The CNN-LSTM hybrid model reached 81.51% test accuracy. By treating CNN features as a small sequence, LSTM adds logical structure to hand-shape positions. This improves performance on unseen signers, but since the task recognizes single, static alphabets, the temporal gain remains limited. The model still performs strongly and consistently, demonstrating good class separation and stable predictions.

The EfficientNet-B3 CNN model achieved 78.55% test accuracy, which is slightly lower than the other models. It is lightweight and optimized, making CNN training efficient, but it was more sensitive to signer variation. This means it sometimes confuses letters when different people perform visually similar signs. However, its efficiency makes it a valuable baseline for future research or low-power device deployment.

The ResNet-18 CNN model produced the best and most stable performance, with 81.98% test accuracy. The residual connections inside the backbone help the model preserve fine details like finger gaps and palm posture, even in challenging lighting. As a result, it generalized better across different signers and maintained strong precision and recall, making its real-world inference the most reliable.

B. ResNet-18 Result Analysis

This research follows a single unified preprocessing-to-training pipeline, and all models were trained using CUDA-enabled GPU hardware for fast and stable optimization. The following table summarizes the parameters used for model training. The table-II shows the Model training parameters.

TABLE II
MODEL TRAINING PARAMETERS

Training Parameter	Value
Learning rate (stage-wise)	1e-3 → 5e-4 → 1e-4 → 5e-5
Image dimensions	224 × 224 × 3
Batch size	32
Total epochs (staged)	50
Optimizer	Adam
Detection model	YOLOv8-nano (hand crop)
Train/validation ratio	80% / 20%

The model achieved 81.98% test accuracy, showing good general recognition ability for a 38-class Bengali sign alphabet dataset, where many signs look very similar, making the task naturally difficult. Instead of using an easier user-dependent test, we used a fully signer-independent split with unseen users to test real-world generalization, not memorization. Although ResNet-18 is efficient, the dataset is small and sign-focused compared to ImageNet and contains real capture challenges like lighting changes, blur, scale mismatch, and self-occlusion, yet the model still holds balanced precision and F1 (> 0.82), proving stable learning. A four-stage fine-tuning process with slow unfreezing and learning-rate reduction helped avoid over-training, making the results trustworthy and a strong baseline for future improvements using attention or articulation-based models.

C. Training, Validation and Inference Reliability

This research applied a single unified preprocessing and training pipeline. The dataset was partitioned into 80% training and 20% validation. All training and evaluation were performed using CUDA-enabled GPU hardware for fast, stable convergence. The Fig.5 represents the Training vs Validation loss accuracy.

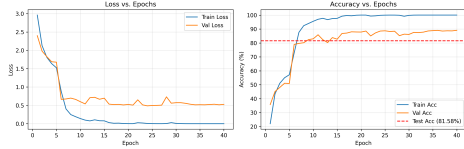


Fig. 5. Training vs. validation loss and accuracy

The training loss decreases consistently, meaning the model is learning real hand-sign features. The validation loss remains close, indicating the model is not memorizing individual signers. Accuracy increases gradually, showing smooth and stable learning behavior. The small train-validation gap suggests good generalization ability for unseen users.

The Fig.6 represents the Micro and macro ROC-AUC curves.

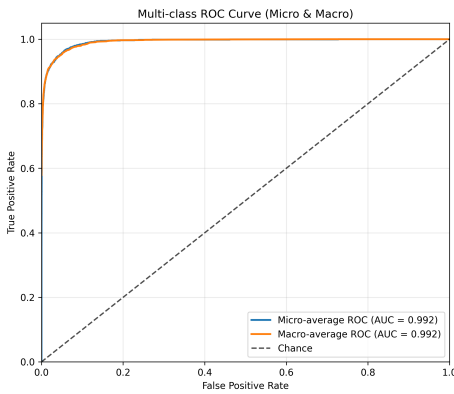


Fig. 6. Micro and macro ROC-AUC curves

The ROC curve stays near the top-left, which means the model can clearly separate between alphabet classes. The AUC score is around 0.992, showing high confidence in predictions. Micro and macro curves behaving similarly indicate consistent classification across all 38 classes. This proves the backbone features are very strong and reliable for inference.

The Fig.7 represents the Confusion matrix on signer-independent test set.

The matrix demonstrates strong signer-independent performance, as most prediction values lie on the diagonal, confirming high correct-class recognition across 38 BdSL alphabet signs. The relatively few off-diagonal entries indicate minimal confusion, mostly between signs that share structural posture similarities, which is a common challenge in static sign-language recognition. The distribution suggests that the model

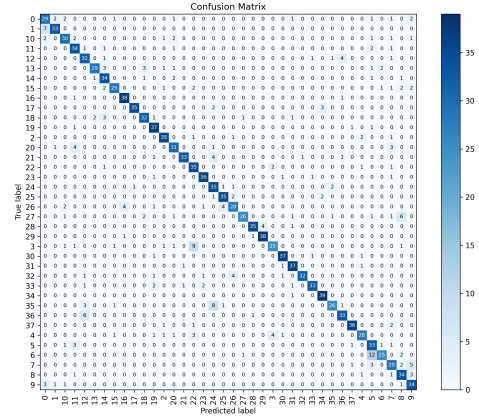


Fig. 7. Confusion matrix on signer-independent test set

learned stable spatial hand features—such as finger articulation and palm orientation—that generalize well to completely unseen users during testing.

Among all evaluated models, ResNet-18 is the best-performing architecture in our experimentation, delivering the highest accuracy and most stable signer-independent predictions.

D. Computational Efficiency Discussion

Beyond recognition accuracy, computational efficiency is very important for the practical deployment of sign language recognition systems, especially in real time and resource related environments. The three architectures show clear differences in parameter complexity, inference cost and training behaviour.

The CNN-LSTM hybrid model shows an LSTM layer to collect sequential information which increases memory consumption and response time during inference. Even after providing limited robustness benefits by temporal modeling, the recurrent layers process data step by step rather than all at once. As a result the model is not suitable for fast and embedded systems, especially for static alphabet recognition tasks where time based data is not very important.

EfficientNet-B3 is mainly designed to give good accuracy using a small amount of parameters and it trains very well. But as the model is highly optimized, it becomes more sensitive to differences between the people who perform the sign language gestures and small spatial changes. Because of this, a slight drop in accuracy has been detected, showing a trade-off between efficiency and robustness.

When Compared with the other two, ResNet-18 achieves the most balanced performance. It's simple and lightweight design, along with residual connections that help store important details, allows the model to train fastly and smoothly and also make predictions very well. It maintains real-time GPU performance while delivering strong recognition accuracy. Overall, ResNet-18 gives the best balance between accuracy and efficiency which makes it a better choice for real-world Bengali sign language recognition and it also gives a strong basic model that can easily be improved in future.

The table-III represents the comparison of our paper’s model with the recent static sign language recognition works.

TABLE III
COMPARISON WITH RECENT 2024–2025 STATIC SIGN LANGUAGE
RECOGNITION WORKS

Ref.	Dataset	Methodology	Acc. (%)
[1] Huang et al., 2024	Leap Motion Hand Dataset	ResNet + LSTM spatial-sequential model	86.25
[2] Shaikh et al., 2025	Leap Motion Hand Dataset	Statistical + spatio-temporal features	86.75
[3] Farid et al., 2024	Bangla static SL dataset	CNN-based static sign classifier	83.00
[4] Our (ResNet-18)	BdSL 38-class image dataset	ResNet-18 with staged fine-tuning	81.98
[5] Our (CNN+LSTM)	BdSL 38-class image dataset	CNN + LSTM sequence classifier	81.51
[6] Our (EfficientNet-B3)	BdSL 38-class image dataset	EfficientNet-B3 transfer learning	78.55

CONCLUSION

In this study, three deep learning architectures were evaluated for 38-class signer-independent Bengali Sign Language alphabet recognition. The CNN-only EfficientNet-B3 model achieved 78.55% test accuracy. Despite nearly perfect training accuracy, the gap with test results shows that it faces more difficulty when recognizing signs from completely unseen users. It remains a strong and efficient spatial-learning baseline but showed comparatively higher class confusion.

The CNN–LSTM hybrid model introduced sequence awareness over spatial features, reaching 81.51% test accuracy. This confirms that an LSTM layer can help refine feature relationships and reduce ambiguity between structurally similar signs. However, since the recognition task is alphabet-level and static, the improvement remains useful but moderate. The best performance was achieved using ResNet-18, which obtained the highest 81.98% test accuracy. ResNet-18’s residual connections preserved fine hand cues such as finger spacing, palm orientation, and knuckle posture, enabling strong spatial learning. The LSTM layer further improved class discrimination by learning dependencies across spatial feature positions. Grad-CAM analysis verified that the model consistently attends to dominant hand regions, supporting interpretable and reliable inference.

Although the model shows strong signer-independent performance, the dataset size and signer diversity can still be expanded further to improve scalability and cross-domain robustness.

Future research should focus on hyper-parameter tuning, ensemble learning, and deeper segmentation, explore sign-aware self-attention or transformer-based sequence modules, evaluate

multi-resolution input scaling, and verify performance through real deployment testing with domain experts.

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